

EFFECT OF THE LOCAL SPECIE OF *ALOE* ON THE HEALING PROCESS

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ABSTRACT

Aloe barbadensis Mill. (Syn. *Aloe vera* Tourn. ex. Linn.) has been used as a healing agent. There are certain reports which indicate that *Aloe barbadensis* helps in the process of healing of wounds. However, no study has been carried out on this plant in Pakistan.

In the present study results show that the cream prepared from the local specie of *Aloe barbadensis* has very significant effect on the process of healing.

INTRODUCTION

Aloe barbadensis is locally known as "Ghikunvar". It is a short- stemmed succulent herb which is widely distributed in Asia, Africa and other tropical parts of the world (Bhatnagar, Fransworth). It has a very ancient history for healing wounds and use in cosmetics (Shousen, Dymock et al.). Its leaves are very useful in the therapy of asthma, gastrointestinal ulcers, burns and chronic wounds (Bhatnagar, Fransworth, Shousen, Dymock, Rubeena et al.). Recent biological studies showed that its leaves possess analgesic, antipyretic, purgative, anticancer (Fransworth, Rubeena et al.), antimicrobial (Patel et al.) and immunological adjuvant (Hart et al.) activities. Chemical constituents commonly found in *A. barbadensis* are of anthraquinones (barbaloin), chromones (aloesin) and polysaccharides (glucomannans) (Hart et al., Manitto et al., Axing et al., McAnalley et al.), which are also responsible for its antiviral, immunological adjuvant and wound healing properties (Hart et al., McAnalley et al., Sydiskis et al.).

The wound healing activity of *Aloe barbadensis* has been investigated in rats. Wound healing is a complex (but orderly) phenomenon involving a number of well orchestrated processes, including regeneration of parenchymal cells, migration and proliferation of both parenchymal and connective tissue cells, synthesis of ECM proteins, remodeling of connective tissue and parenchymal components, and collagenization and acquisition of wound strength.

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MATERIALS AND METHODS

1. PHC-012 Cream Prepared by Dr. H.M.I. Institute of Pharmacology & Herbal Sciences.
2. Rats (Sprague-Dawley).
3. Chemical Ingredients used in cream preparation.
4. *Aloe barbadensis* (Leaves)

Table 1: Wound healing activity of *Aloe barbadensis*

Treatment	Wound area in mm ² with percentage of wound healing in brackets				No. of days for complete contraction of wound
	Days				
	0	6	9	13	
Cream (<i>Aloe barbadensis</i>)	590 (0.00%)	510 (13.55%)	61 (89.66%)	10 (98.30%)	13 days
White soft Paraffin	711 (0.00%)	484 (31.94%)	193 (72.86%)	77 (89.18%)	20 days

Experimental: (Wound Healing Effect)

Healthy rats of both sexes (200-300 gm) served as experimental models. A wound of about 20 mm square area was created on both sides of the spinal cord of rats after removing the hairs (Balsam et al., Ahmed et al., Khanna et al.). Due care was taken to maintain the rats in the same experimental conditions in order to have a direct bearing on the healing process. The animals were given food and water ad libitum.

The cream of *Aloe barbadensis* were applied daily for 14 days (Chopra et al., Carter) on the wounds of the different sets of rats till there was complete contraction. An identical set of experiment on the same rats using white soft paraffin was performed.

RESULTS AND DISCUSSION

The cream which is prepared from the local specie of *Aloe barbadensis* cultivated in the garden of Bait al-Hikmah, has been found to enhance healing by influencing the following three parameters i.e. Wound contraction, Epithelialisation and Granulation tissue formation (Table 1). The wound contraction time has been found to be 13 days. The animals showed wavy pattern and a wrinkled appearance along the edges of the wound.

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